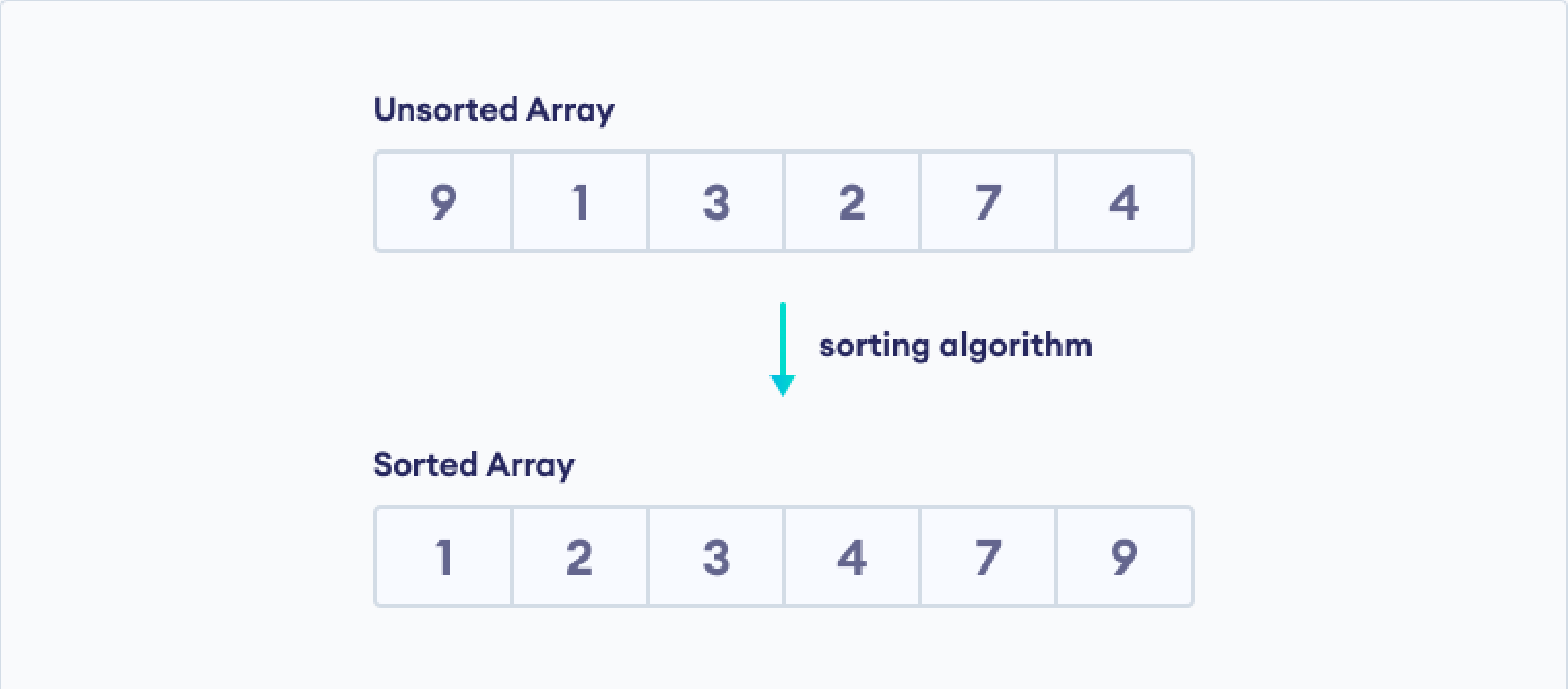
Sorting

Program Studi Pendidikan Komputer
Fakultas Keguruan dan Ilmu Pendidikan
Universitas Lambung Mangkurat

Sorting

Algoritma pengurutan (*sorting algorithm*) pengurutan digunakan untuk mengatur elemen-elemen dalam sebuah array/list menjadi urutan tertentu.



Sorting an array

Sorting

Sorting Algorithms:

- Bubble Sort
- Selection Sort
- Insertion Sort
- Merge Sort
- Quicksort
- Counting Sort
- Radix Sort
- Bucket Sort
- Heap Sort
- Shell Sort

Sorting

Complexity of Sorting Algorithms

The efficiency of any sorting algorithm is determined by the **time complexity** and **space complexity** of the algorithm.

- **Time Complexity:** Time complexity refers to the time taken by an algorithm to complete its execution with respect to the size of the input.

- **Space Complexity:** Space complexity refers to the total amount of memory used by the algorithm for a complete execution.

Complexity Analysis

Sorting Algorithm	Time Complexity - Best	Time Complexity - Worst	Time Complexity - Average	Space Complexity
Bubble Sort	n	n^2	n^2	1
Selection Sort	n^2	n^2	n^2	1
Insertion Sort	n	n^2	n^2	1
Merge Sort	nlog n	nlog n	nlog n	n
Quicksort	nlog n	n^2	nlog n	log n
Counting Sort	n+k	n+k	n+k	max
Radix Sort	n+k	n+k	n+k	max
Bucket Sort	n+k	n^2	n	n+k
Heap Sort	nlog n	nlog n	nlog n	1
Shell Sort	nlog n	n^2	nlog n	1

Bubble Sort

3	2	4	1

Outer Loop: i = 0 to n-1 *Dari item pertama ke item terkahir

Outer Loop: j = 0 to n-1-i *karena item terakhir sudah terurut

Bubble Sort

```
def bubblesort(mylist):
    for i in range (0, len(mylist)-1):
        for j in range(0, len(mylist)-1-i):
            if mylist[j]>mylist[j+1]:
                mylist[j],mylist[j+1] = mylist[j+1],mylist[j]
    return mylist

theList = [3,2,4,1]
print(bubblesort(theList))
```

Bubble Sort

```
def bubble(list1):
    indexing_length = len(list1)-1
    sorted = False

    while not sorted:
        sorted = True
        for i in range(0, indexing_length):
            if list1[i]>list1[i+1]:
                sorted = False
                list1[i],list1[i+1] = list1[i+1],list1[i]
    return list1

print(bubble([3,2,4,1]))
```

Terima Kasih